

Case Study Report

Bike Sharing Analysis

Executive Summary

This case study analyzes Cyclistic bike-share usage in Chicago to identify differences between casual riders and annual members and to identify potential opportunities for increasing casual riders' conversion to annual memberships. The analysis uses 5,932,349 rides covering the 12-month period from July 2025 through June 2026. The data were combined and processed in Microsoft Access, summarized for analysis, and visualized using Tableau.

The results show several clear behavioral differences between casual riders and annual members. Casual riders accounted for 2,116,634 rides, or 35.7% of all rides, while annual members accounted for 64.3%. The most substantial difference was ride duration: casual riders averaged 21.73 minutes per ride compared with 12.46 minutes for members. In addition, 26.03% of casual rides lasted at least 20 minutes, compared with 14.16% of member rides, indicating that casual riders have a considerably larger share of medium- and longer-duration trips.

Casual riders also showed different temporal and spatial patterns. Weekend riding was more important among casual riders, with Saturday and Sunday together accounting for 37.53% of casual rides compared with 23.31% of member rides. Casual riders also showed relatively stronger afternoon and evening activity, while members had stronger morning and evening peak patterns. The station analysis showed that casual activity was more concentrated around several lakefront, recreational, cultural, and destination-oriented locations. Bike type, in contrast, showed a relatively small difference between the two groups, with both groups predominantly using electric bikes.

Based on these behavioral patterns, several potential marketing opportunities were identified. Cyclistic could consider emphasizing membership value among casual riders who regularly take longer rides, timing membership communications around periods of high casual activity, and using stations with high casual-rider shares as potential locations for membership messaging. Marketing communications could also be tailored to different behavioral contexts rather than treating all casual riders as a single segment.

These findings should be interpreted as descriptive associations rather than evidence that a particular marketing strategy will cause membership conversion. The dataset does not provide a stable individual rider identifier and therefore cannot track individual casual riders over time. The recommended next step is to test targeted marketing approaches using appropriate comparison groups and evaluate outcomes through measures such as membership sign-ups, conversion rates, repeat usage, and campaign engagement.

1. Ask

Business Task

Cyclistic is a bike-share company in Chicago with two main types of riders: **casual riders**, who purchase individual rides, and **annual members**, who purchase annual memberships. The company wants to increase the number of casual riders who become annual members.

The marketing team wants to understand how casual riders and annual members use Cyclistic bikes differently. By identifying differences in riding patterns, the analysis can help identify potential opportunities for targeted marketing strategies aimed at casual riders.

Business Question

How do casual riders and annual members use Cyclistic bikes differently, and what marketing opportunities can be identified from these differences to encourage casual riders to become annual members?

Key Questions

The analysis examines differences between the two rider groups in terms of:

- Ride duration
- Ride-duration distribution
- Day of the week
- Time of day
- Monthly and seasonal patterns
- Bike type

- Starting locations

The findings from these comparisons will be used to identify patterns among casual riders that may inform membership-focused marketing strategies.

2. Prepare

The analysis used Cyclistic bike-share trip data covering a 12-month period from **July 2025 through June 2026**. The data consisted of 12 monthly trip-data files, with each file representing one month of Cyclistic bike-share activity in Chicago.

The dataset contained information about individual bike trips, including rider type, ride start and end times, start and end stations, start and end station IDs, and bike type. The main variables used in this analysis were **rider type, ride duration, day of the week, hour of the day, month, bike type, and starting station**.

The 12 monthly datasets were imported into **Microsoft Access** and combined using a Union query to create a single dataset for analysis. The combined dataset contained **5,932,349 rides**, including **2,116,634 casual rides** and **3,815,715 member rides**.

Before analysis, the trip data were organized to allow comparisons between casual riders and annual members. Ride duration was calculated from the recorded start and end times. Additional variables, including ride hour, day of week, and month, were used to examine differences in riding patterns over time. Station-level analysis used station IDs where available to reduce problems caused by variations in station names.

The analysis focused on behavioral patterns that could be compared between the two rider groups. The prepared data were subsequently summarized in Microsoft Access and the resulting analysis tables were exported to Excel for visualization in Tableau Desktop.

Data Limitations

Several limitations were considered during the preparation and analysis of the data. Some trips contained missing station information, which limited the use of station-level information for some observations. In addition, the dataset identifies riders by membership type but does not provide a stable individual rider identifier. Therefore, the analysis describes **aggregate riding patterns** rather than the behavior of individual casual riders.

The analysis also identifies associations and patterns in riding behavior; it does not establish the reasons why riders behave in a particular way or demonstrate that a specific marketing strategy will cause casual riders to purchase memberships.

3. Process

The raw monthly trip datasets were processed in Microsoft Access before being used for analysis and visualization. The 12 monthly files from July 2025 through June 2026 were imported into separate tables within an Access database. Each monthly table contained the trip-level records for that month.

The monthly tables were then combined using a **Union query** to create one consolidated dataset covering the entire 12-month study period. The resulting dataset contained **5,932,349 individual ride records**.

Several calculated variables were created to support the analysis. **Ride duration** was calculated from the difference between the ride start and end times and expressed in minutes. The start-time information was also used to derive the **hour of the day**, **day of the week**, and **month** in which each ride began. These variables allowed the data to be analyzed according to different temporal patterns.

The data were then grouped according to **rider type**, distinguishing between casual riders and annual members. Aggregations were performed to calculate measures such as total rides, average ride duration, and the number or proportion of rides across different days, hours, months, bike types, and starting stations.

For the ride-duration analysis, rides were additionally divided into five duration groups:

- Under 10 minutes
- 10–19 minutes
- 20–29 minutes
- 30–59 minutes
- 60 minutes or more

Station-level analysis was conducted using station IDs as the primary identifier where available. This helped reduce inconsistencies that could occur when the same station appeared under slightly different names. Station-level summaries were used

to examine where casual riders were particularly concentrated and to compare casual activity across starting locations.

After the required calculations and aggregations were completed in Access, the resulting summary tables were exported to Excel. These summarized datasets were then connected to **Tableau Desktop** for visualization. Using summarized tables rather than the full trip-level dataset in Tableau reduced the computational burden while preserving the information required for the analysis.

The processed data were checked for consistency before visualization. The total number of rides and the number of rides by rider type were verified against the consolidated dataset. The resulting analytical tables were then used to construct the charts presented in the final dashboard.

4. Analyze

The analysis compared casual riders and annual members across several dimensions of Cyclistic usage: overall ride shares, ride duration, day of the week, time of day, monthly patterns, bike type, and station location. The objective was to identify the behavioral differences that are most relevant to understanding potential opportunities for membership marketing.

Overall Membership Share

Between July 2025 and June 2026, Cyclistic recorded **5,932,349 rides**. Casual riders accounted for **2,116,634 rides (35.7%)**, while members accounted for **3,815,715 rides (64.3%)**. Thus, approximately one-third of all rides were made by casual riders and two-thirds by members.

Although members generated the majority of rides, the volume of casual riding was substantial enough to represent an important segment for the analysis.

Ride Duration

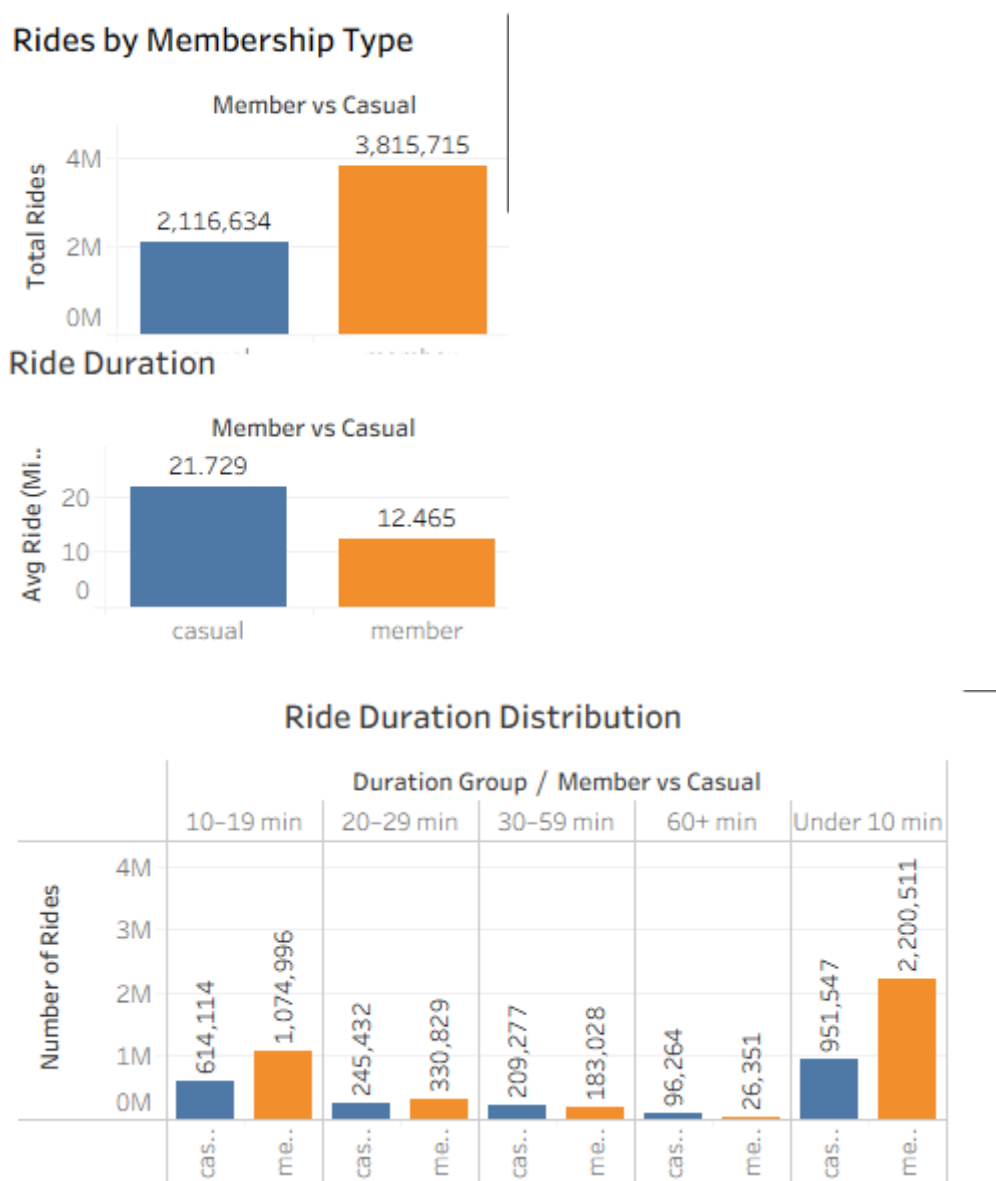
One of the clearest differences between the two groups was ride duration. Casual riders had an average ride duration of **21.73 minutes**, compared with **12.46 minutes** for members—a difference of **9.26 minutes**. The analysis found that casual rides were approximately **74% longer on average** than member rides.

The distribution of ride duration provides additional evidence of this difference. Among casual rides, **44.96%** lasted under 10 minutes, compared with **57.67%** of

member rides. At the longer end of the distribution, **9.89%** of casual rides lasted 30–59 minutes compared with **4.80%** of member rides, while **4.55%** of casual rides lasted 60 minutes or more compared with only **0.69%** of member rides.

Combining the duration categories shows an especially clear distinction: **26.03% of casual rides lasted 20 minutes or longer, compared with 14.16% of member rides.** For rides lasting 30 minutes or longer, the corresponding figures were **14.44% for casual riders and 5.49% for members.**

Therefore, the difference is not simply a result of the average duration. Casual riders have a substantially larger share of medium- and long-duration rides.



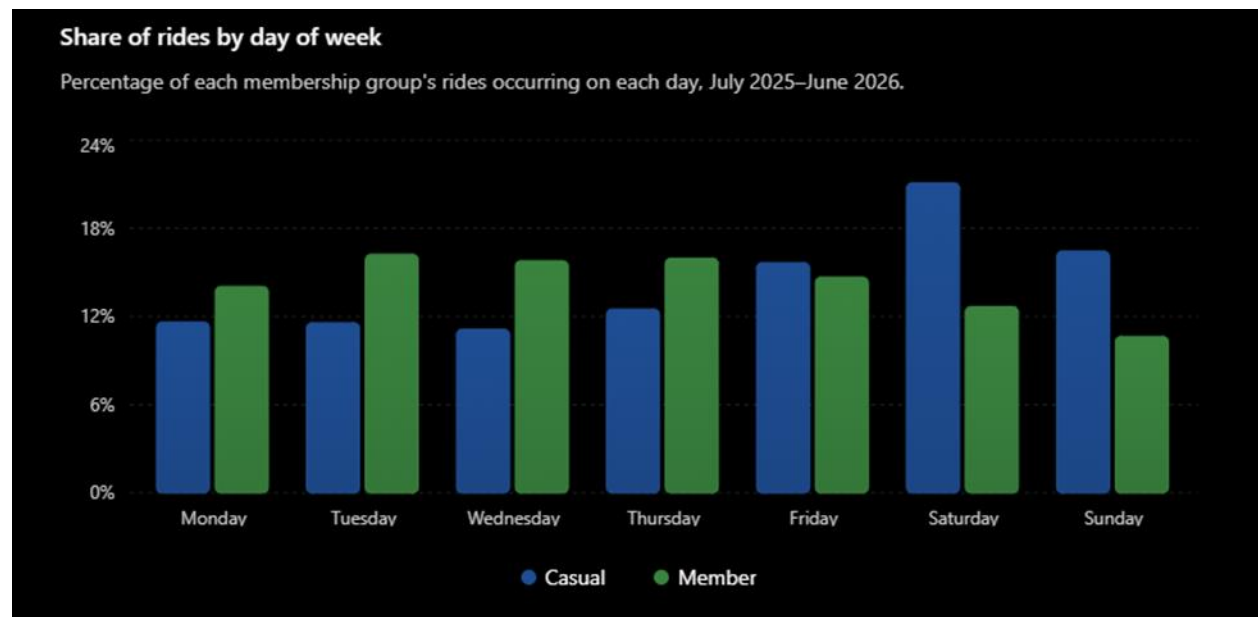
Day of the Week

The two groups also displayed different weekly riding patterns. Casual riders were more concentrated toward the weekend, while members had relatively greater weekday usage.

When each group's rides were considered as a percentage of its own total, **Saturday accounted for 21.09% of casual rides and 12.67% of member rides**, while Sunday accounted for **16.44% of casual rides and 10.64% of member rides**. Together, Saturday and Sunday represented **37.53% of casual rides**, compared with **23.31% of member rides**.

A second measure reinforces this pattern. Casual riders accounted for **48.00% of all Cyclistic rides on Saturday** and **46.16% on Sunday**, compared with approximately **28.1%–31.5% on Monday through Thursday**. Friday was intermediate at **37.18%**.

These two measures should be distinguished: the first describes how each group's own rides are distributed across the week, while the second describes the casual share of all rides occurring on each day. Together, they show that weekend activity is particularly important in the casual-rider segment.



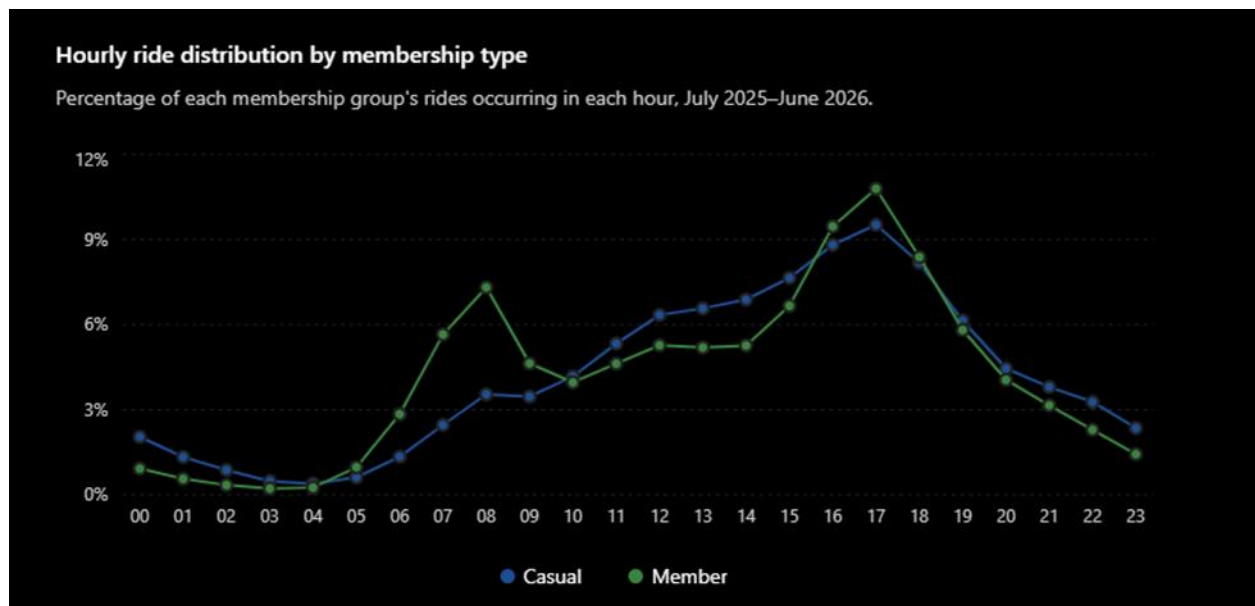
Time of Day

The hourly analysis revealed another clear difference. Members showed a stronger concentration during morning and evening peak periods, while casual usage increased through the day and remained relatively strong into the evening.

Between **06:00 and 09:00**, only **10.80% of casual rides** occurred, compared with **20.43% of member rides**. In contrast, between **20:00 and 23:00**, casual riders accounted for **13.87%** of their rides compared with **10.92%** for members. During **16:00–18:00**, the shares were relatively close at **26.51% for casual riders and 28.63% for members**.

Looking at casual riders' share of all rides by hour provides another useful perspective. Casual riders represented only **19.48%–21.16% of rides between 06:00 and 08:00**, but their share increased to **42.07% at 14:00**. It also increased again during the late evening, reaching **44.19% at 22:00 and 47.73% at 23:00**.

The overnight period also had a high casual share, but the absolute number of rides during those hours was relatively low. Therefore, the more meaningful pattern is the combination of relatively low casual representation during the early morning, higher representation during much of the afternoon, and increasing casual share during the late evening.



Monthly and Seasonal Patterns

The monthly analysis shows a strong seasonal pattern in Cyclistic usage, with both casual and member rides varying substantially across the 12-month study period.

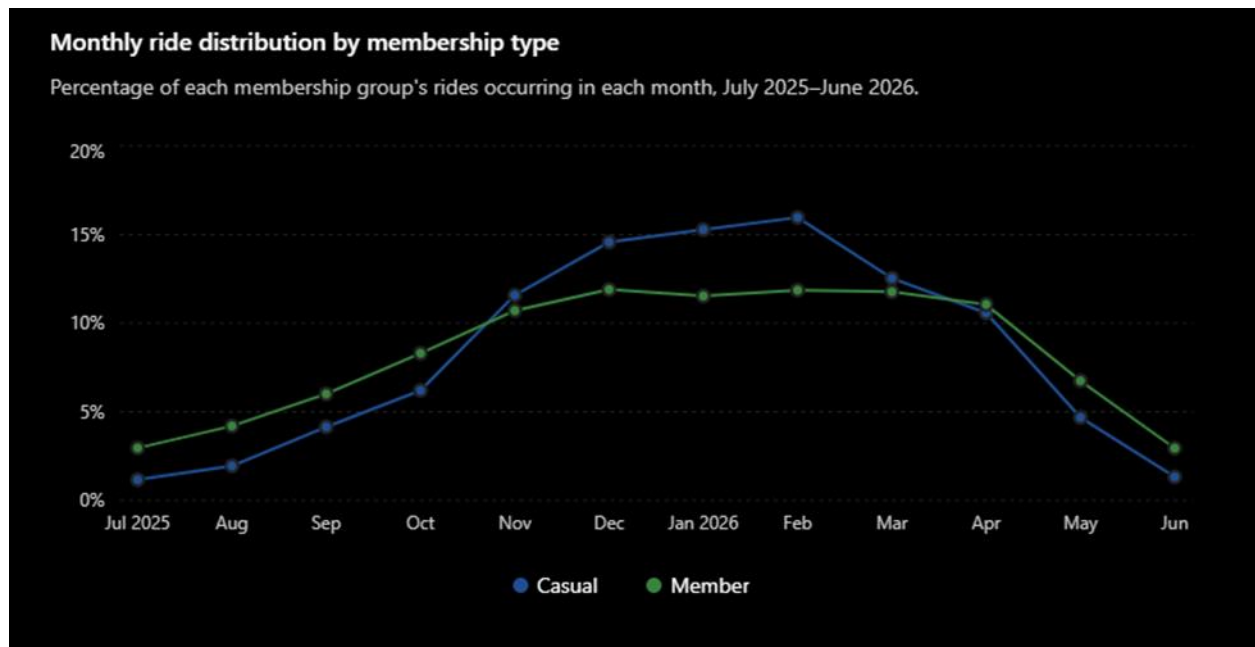
Casual rides were highest during the warmer months and declined considerably during the winter period.

Casual rides totaled 323,352 in July 2025 and increased to a peak of 337,878 in August 2025. Activity then declined to 265,268 in September and 224,038 in October. The decline became more pronounced in November and December, when casual rides fell to 99,091 and 28,090 respectively. Casual activity reached its lowest level in January 2026 at 24,740 rides before gradually increasing through the following months, reaching 41,173 in February, 87,850 in March, 131,526 in April, 245,143 in May, and 308,485 in June.

Member rides followed a similar broad seasonal pattern but remained higher than casual rides throughout the study period. Member rides increased from 440,106 in July 2025 to 452,439 in August, remained relatively high through September and October, and then declined substantially during November through January. Member activity subsequently increased from February through June 2026, reaching 454,089 rides in June.

The results therefore indicate that riding activity is strongly influenced by the time of year. Casual riders show particularly large changes in monthly ride volume, with the highest activity occurring during the warmer months and the lowest activity occurring during winter. This pattern suggests that the timing of membership marketing may be important, particularly during periods when casual riding activity is high and there is greater potential exposure to casual users.

However, the monthly analysis describes when riding activity occurred and does not establish the reasons for the observed seasonal pattern or whether seasonal marketing campaigns would increase membership conversion.



Bike Type

Both groups predominantly used electric bikes. Among casual riders, **70.00% of rides were made using electric bikes and 30.00% using classic bikes**. Among members, **66.32% were electric and 33.68% were classic**. The difference in electric-bike share was only **3.68 percentage points**.

This means that bike type does not distinguish the two groups as strongly as ride duration, day of week, time of day, or season. Both groups showed a similar overall preference for electric bikes.

However, when bike type and ride duration were examined together, the duration difference between casual and member riders remained for both bike types. For classic bikes, average duration was **39.62 minutes for casual riders versus 15.18 minutes for members**. For electric bikes, the averages were **14.06 minutes for casual riders and 11.09 minutes for members**.

This indicates that the longer average duration of casual rides cannot simply be attributed to their greater use of electric bikes. The difference is particularly pronounced for classic-bike rides.

Station and Location Patterns

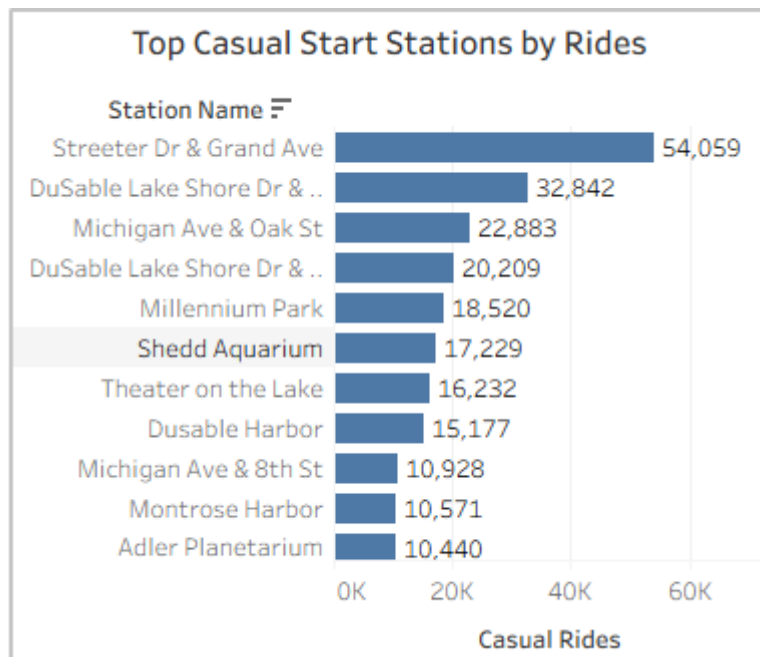
Station-level analysis showed a different spatial pattern between the two groups. Several of the most frequently used casual starting stations were located around lakefront, recreational, cultural, or major attraction-oriented areas. Examples include **Navy Pier, DuSable Lake Shore Dr & Monroe St, Millennium Park, Shedd Aquarium, Theater on the Lake, Dusable Harbor, and Montrose Harbor.**

Casual riding was also more concentrated among its highest-use stations. Among rides with available starting-station information, the top 10 casual starting stations accounted for approximately **9.42% of all casual rides**, compared with **5.08% for the top 10 member starting stations**. The equivalent figures for ending stations were **9.86% for casual riders and 5.08% for members**.

Several individual stations had particularly high casual shares. For example, **Shedd Aquarium had an 81.98% casual share**, Field Museum had **81.09%**, DuSable Lake Shore Dr & Monroe St had **78.97%**, and Streeter Dr & Grand Ave had **77.50%**.

The similarity between casual starting and ending stations also strengthens the spatial finding. Nine of the top 10 casual starting stations also appeared among the top 10 casual ending stations, and several retained very similar rankings.

The station data, however, do not directly identify the purpose of an individual trip. Therefore, the analysis does not establish that casual riders are tourists. Instead, the evidence shows that casual riding activity is concentrated around several lakefront, recreational, cultural, and visitor-oriented locations.



Overall Behavioral Pattern

Taken together, the analysis reveals a consistent distinction between the two rider groups. Casual riders tend to take **longer rides**, have a **greater concentration of trips on weekends**, show stronger **afternoon and evening usage**, and exhibit greater **seasonal concentration**. Their most frequently used stations are also more concentrated around **lakefront, recreational, and major attraction-oriented locations**.

Members, in contrast, tend to take **shorter rides**, have relatively greater **weekday usage**, show stronger **morning and evening peaks**, have a more even distribution of rides across the year, and use a less concentrated set of high-use stations that are predominantly located around downtown intersections.

The strongest behavioral differences are therefore found in **ride duration, day of week, time of day, seasonal concentration, and station location**. Bike type provides useful descriptive information, but the relatively small difference in electric-bike usage makes it a less important distinguishing characteristic for the business question.

5. Share

The findings from the analysis were communicated through a Tableau dashboard titled **Rider Behavior Dashboard**. The dashboard was designed to present the main differences between casual riders and annual members in a clear and accessible format.

The dashboard brings together visualizations covering rider membership, ride duration, day of the week, time of day, monthly patterns, bike type, and station-level activity. These visualizations allow the two rider groups to be compared across several dimensions of riding behavior.

The dashboard includes the following key visualizations:

- **Total Rides by Rider Type**, showing the overall distribution of rides between casual riders and annual members.
- **Average Ride Duration by Rider Type**, highlighting the difference in average ride length.
- **Rides by Day of Week**, showing the different weekly patterns of casual riders and members.
- **Rides by Hour**, showing differences in the timing of rides throughout the day.
- **Rides by Month**, illustrating the different patterns of riding activity across the 12-month period.
- **Rides by Bike Type**, comparing the use of electric and classic bikes between the two rider groups.
- **Top Casual Start Stations by Casual Share**, identifying locations where casual riders account for a particularly large proportion of rides.
- **Top Casual Start Stations by Total Rides**, identifying locations with high volumes of casual activity.
- **Casual Share by Day of Week** and **Casual Share by Hour**, providing additional views of when casual riders represent a larger proportion of total rides.
- **Average Duration by Bike Type and Rider Type**, showing how ride duration differs across both rider type and bike type.

- **Ride Duration Distribution by Rider Type**, showing the proportion of rides within different duration categories.

The dashboard was organized to make the major behavioral differences visible without requiring the audience to examine the underlying trip-level data. In particular, the visualizations emphasize the longer ride durations, stronger weekend concentration, different time-of-day patterns, seasonal variation, and concentration of casual activity at certain lakefront and destination-oriented stations identified during the analysis.

The dashboard therefore serves as a visual summary of the analysis and provides the basis for communicating the findings to the marketing team. The main insights presented in the dashboard can subsequently be used to develop marketing recommendations aimed at encouraging casual riders to consider annual membership.

6. Act

The analysis shows several clear differences between casual riders and annual members that can inform Cyclistic's marketing strategy. Casual riders account for 35.7% of all rides and generally take longer rides than members, with an average duration of 21.73 minutes compared with 12.46 minutes for members. Casual riders also have a greater concentration of rides on weekends, stronger activity during afternoon and evening periods, and greater seasonal variation. In addition, casual rides are more concentrated at several lakefront, recreational, cultural, and destination-oriented stations.

Based on these findings, Cyclistic could consider the following marketing actions.

a. Target casual riders with longer-ride membership messaging

Ride duration is one of the clearest differences between the two groups. Casual riders average 21.73 minutes per ride, compared with 12.46 minutes for members. In addition, 26.03% of casual rides last at least 20 minutes, compared with 14.16% of member rides, while 14.44% of casual rides last at least 30 minutes compared with 5.49% of member rides. Cyclistic could therefore develop membership messaging that emphasizes the potential value and convenience of membership for riders who regularly take longer

trips. This messaging could be particularly relevant to casual riders whose usage patterns already involve relatively long rides.

b. Focus membership campaigns on weekends and leisure-oriented periods

Casual riders show a substantially stronger weekend pattern than members. Saturday and Sunday together account for 37.53% of casual rides but only 23.31% of member rides. Casual riders also have a higher share of rides during afternoon and evening periods.

Cyclistic could use this pattern to time membership promotions around weekends and other periods when casual riding is more concentrated. Rather than relying only on weekday-oriented messaging, campaigns could be designed around the types of riding periods in which casual users are already particularly active.

c. Use high-casual-share stations as potential marketing locations

The station analysis identifies several locations where casual riders account for a particularly large share of rides. For example, casual riders represent approximately 82% of rides at Shedd Aquarium and approximately 81% at Field Museum. Other high-casual-share locations include DuSable Lake Shore Drive & Monroe, Streeter Drive & Grand, Dusable Harbor, Adler Planetarium, McCormick Place, and Millennium Park.

These locations could provide potential opportunities for targeted membership communication. Cyclistic could consider displaying membership information or promotional messaging at stations with high casual-rider shares, particularly where casual activity is concentrated around lakefront, recreational, cultural, and destination-oriented locations.

d. Consider seasonal timing when planning campaigns

Casual riding varies more substantially across the year than member riding. Consider increasing membership-focused marketing exposure during periods of high casual riding activity, particularly July–October and May–June, when casual riders account for substantially more rides than during the winter months.

This suggests that membership marketing does not necessarily need to be distributed uniformly throughout the year. Cyclistic could use periods of relatively high casual activity as opportunities to increase exposure to membership information and test whether riders who are already using the service frequently respond to membership offers.

e. Develop targeted messaging rather than treating all casual riders as one group

The analysis shows that casual riders are not defined by a single behavior. Their differences from members appear across ride duration, day of week, time of day, seasonality, and station location. Therefore, Cyclistic could consider using different messages for different behavioral contexts rather than applying one identical message to every casual rider.

For example, messaging displayed at high-casual-share recreational locations could emphasize membership for repeated leisure use, while communications aimed at casual riders with longer rides could focus on the value and convenience of membership for extended trips. These approaches are potential marketing applications of the observed patterns rather than conclusions that the analysis has established as causally effective.

Recommendation

Before implementing a large-scale membership campaign, Cyclistic could test these approaches using targeted campaigns and compare their results with appropriate comparison groups. Measures such as membership sign-ups, conversion rates, repeat usage, and campaign engagement could then be monitored to determine whether the proposed strategies are associated with increased conversion of casual riders to annual members.

Recommendation	KPI
Longer-ride membership messaging	Membership conversion rate
Weekend campaigns	Weekend conversion rate
Station-based campaigns	Sign-ups by station
Seasonal campaigns	Conversion by month
Targeted messaging	Conversion by customer segment

Overall, the analysis suggests that the most relevant marketing opportunities are concentrated around the **duration, timing, seasonality, and locations of casual riding**. These findings provide a data-informed basis for targeting membership

communication while recognizing that further testing would be required to determine which specific marketing approaches are most effective.

Conclusion

This analysis examined how casual riders and annual members use Cyclistic bikes differently using trip data from July 2025 through June 2026. The results show clear differences in riding behavior between the two groups. Casual riders generally take longer rides, have a greater concentration of rides on weekends, show stronger afternoon and evening activity, and exhibit greater seasonal variation. Their rides are also more concentrated at certain lakefront, recreational, cultural, and destination-oriented stations. In contrast, annual members show shorter ride durations, greater weekday activity, stronger morning and evening patterns, and a more consistent distribution of rides across the year.

These differences suggest several potential opportunities for Cyclistic to target casual riders with membership-focused marketing. In particular, membership communication could be tailored around longer rides, weekend and leisure-oriented usage, periods of high casual activity, and locations where casual riders represent a large share of rides. However, the analysis identifies behavioral associations rather than the reasons behind them, and it does not establish that any particular marketing strategy will cause casual riders to become members.

Overall, the analysis provides a data-informed basis for developing and testing targeted membership campaigns. Cyclistic can use the identified behavioral patterns to design targeted marketing initiatives and then evaluate their effectiveness using measures such as membership conversion, repeat usage, and campaign engagement.

Appendix

Appendix A — Detailed Analysis Tables

Appendix B — [Tableau Dashboard](#)

Appendix C — Data Processing Notes